DISCRETE TRANSISTOR OP-AMP

Jason Gallicchio and Claude, rev A, 2026-10-24

https://github.com/gallicchio/op-amp-pcb
circuit, scripts, schematic, this board

The op-amp of Learning the Art of Electronics
(Hayes & Horowitz, 2016), built from six
2N3904 / 2N3906 so you can probe every node.

POWER: +15 V, GND, −15 V into the terminal (±16 V max), 20 mA.
  Set the supply's current limit to 50 mA.
  Both LEDs light if it is right.
SWITCHES, usual positions:
  TAIL 1 on: the Q6 current source.
  COMPENSATION 1 on: 100 pF + 1k.
  FEEDBACK as needed (below).
  TAIL 2 instead: a 10k tail resistor.
  COMPENSATION 2 instead: the bare 100 pF, no 1k (the book's);
    the x11 amp still works, a follower rings.
FEEDBACK:
  1 = X to IN−: gain 11.
  2 = OUT to IN−: follower.
  BOTH OFF: a bare op-amp.
AS AN OP-AMP IN YOUR OWN CIRCUIT (Labs 6–9):
  FEEDBACK both off.
  IN−, IN+, OUT and GND on the terminals are pins 2, 3, 6 of a '411.
  Mind its 2 uA of input bias current
    (10 M feedback for photodiode will not work; everything else should).
  A short at OUT draws about 45 mA, limited by R17 (not in the book).
OFFSET NULL: as a follower with IN+ grounded, turn RV1 until OUT reads 0 V.
INPUTS: ±12 V.
OUT: ±10 V into 1k, through R17's 330 ohm, inside the loop.
  A short to ground is survived; to a rail, only with the 50 mA limit.

Discrete Transistor Op-Amp rev A, bottom, seen from below (mirrored): silkscreen, scale 1.90:1